Index laws

1

a 2^{21} **b** 22^{8} **c** 11^{4} **d** 7^{5}

e 5^{48} **f** 15^{4} **g** 23^{79} **h** 17^{8}

i 19^{5} **j** 12^{7} **k** 13^{8} **l** 15^{27}

2

a 8 **b** 7 **c** 34 **d** 5

e 3 **f** 6 **g** 5 **h** 38

i 5 **j** 2

3

a 1 679 616 **b** 9 261 **c** 1 024 **d** 390 625

e 16 807 **f** 4 096 **g** 121 **h** 729

i 36 **j** 1 **k** 1 728 **l** 36

Negative bases

4

a $(-11)^{13}$ **b** $(-21)^{8}$ **c** -15^{2} **d** $(-3)^{7}$

e -12 **f** $(-30)^{3}$ **g** 8^{3} **h** 11^{11}

i $(-7)^{5}$ **j** $(-6)^{2}$

5

a -7 **b** -3 **c** 50 **d** 2

e 49 **f** 12 **g** -2 **h** -21

i 3 **j** -55

6

a 256 **b** 64 **c** -243 **d** $-8\,000$

e -27 **f** $-3\,125$ **g** -216 **h** $-1\,977\,326\,743$

i 1 225 **j** 531 441 **k** 64 **l** $-8\,000$

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Fractional bases

7

a $\frac{1}{3^4}$ **b** $\frac{3^3}{8^3}$ **c** $\frac{1^8}{4^8}$ **d** $\frac{5^2}{2^2}$

e $\frac{10^5}{33^5}$ **f** $\frac{2^6}{35^6}$ **g** $\frac{5^3}{18^3}$ **h** $\frac{29^7}{41^7}$

i $\frac{11^9}{13^9}$ **j** $\frac{20^2}{3^2}$ **k** $\frac{17^4}{4^4}$ **l** $\frac{31^5}{50^5}$

8

a 3 **b** 3 **c** 2 **d** 1

e 3 **f** 16 **g** 9 **h** 625

i 2 **j** 121 **k** 5 **l** 5

9

a $\frac{1}{9}$ **b** $\frac{27}{125}$ **c** $\frac{1}{32}$ **d** $\frac{49}{121}$

e $\frac{27}{512}$ **f** $\frac{144}{169}$ **g** $\frac{125}{729}$ **h** $\frac{16}{81}$

i $\frac{4\,096}{15\,625}$ **j** $\frac{8}{125}$ **k** $\frac{25}{49}$ **l** $\frac{1\,000}{2\,744}$

10

a No **b** Yes **c** No **d** Yes

Negative indices

11 Less than 1.

12

a $\frac{1}{a^9}$ **b** a^n **c** $\frac{1}{4a^9}$ **d** $\frac{b^m}{a^n}$

e $\frac{1}{p^2}$ **f** $\frac{3}{x^4}$ **g** $\frac{7}{x^9}$ **h** $\frac{q^3}{p^2}$

i $\frac{8}{p^3}$ **j** $\frac{2y^3}{x^8}$ **k** $\frac{c^3}{6b^7}$ **l** $\frac{p^4}{m^5n^4}$

m $\frac{3a^4}{b^5c^7}$ **n** $\frac{l^7}{16k^3}$ **o** $\frac{g^2}{f^7h^8}$ **p** $\frac{s^4u^6}{5t^9}$

13

a u^{-4} **b** $2r^{-6}$ **c** xy^{-2} **d** $4p^{-6}q^{-7}$

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a $20y^6$ b $28a^2$ c $\frac{-15}{x^4}$ d $\frac{12}{y^5}$

e $8h^7$ f $\frac{6}{y^7}$ g $\frac{16}{y^3}$ h $25m^3$

i $\frac{3}{x^7}$ j $\frac{3}{x^4}$ k $3x^8$ l $\frac{3}{2c^2}$

m $\frac{5h^9}{4}$ n $3x^8$ o $\frac{3}{2p}$ p $\frac{9}{4w^{10}}$

15

a $\frac{1}{8m^3}$ b $\frac{256}{m^{24}}$ c $\frac{p^8}{9}$ d $\frac{1}{9y^4}$

e $\frac{64}{y^3}$ f $\frac{y^4}{x^5}$ g $\frac{y^{36}}{x^{28}}$ h $5x^5$

i $\frac{5}{x^4}$ j $\frac{81}{z^4}$ k $\frac{q^7}{p^3}$ l $\frac{x^8}{y^{16}}$

16

a $n = -2$ b $n = -3$ c $n = -4$ d $n = -3$

17 $\left(x^4y^{-3}\right)^{-4} = \frac{y^{12}}{x^{16}}$

18 The -1 index only applies to the term directly in front of it. In this case it only applies to the a, not the 5. The correct working is $5a^{-1} = \frac{5}{a}$.

19

a $32y^8$ b $60y^5$

20

a $\frac{10z^9}{21x^3y^7}$ b $\frac{160y^6}{27x}$

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